Week 9 Notes: Circuit-Level Gate Noise on the VQE

Integration, Verification, and Depth Scaling

June 24, 2026

1 Purpose and Context

Week 7 produced a *validated* parity-constrained VQE: across the chemical-potential sweep it returns optimal parameters $\theta^*(\mu)$ that reproduce the even-parity ground state, benchmarked against exact diagonalization for energy, parity, subspace fidelity, and the edge string $O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$. Week 7 also ran a *depth diagnostic*: *noiselessly*, the subspace infidelity falls as the number of ansatz repetitions r grows, so a deeper ansatz prepares a better state.

That conclusion is only half the story on real hardware. A NISQ device builds the state with a *circuit*, and error enters *through the gates*: every entangling operation injects a fresh error channel, so the total corruption grows with the gate count and the circuit depth. Week 9 integrates this correctly with Qiskit's noise stack and asks the quantitative question Week 7 could only hint at: as the ansatz deepens, the prepared state improves but the accumulated gate noise worsens — so how does the edge-string diagnostic actually behave?

This note has three goals:

1. **Theory.** Treat a noisy gate as a completely positive trace-preserving (CPTP) map composed with the ideal unitary, and write the depolarizing and thermal-relaxation channels in Kraus/Pauli form.
2. **Practice.** Build a Qiskit `NoiseModel`, attach gate errors to instructions, read the edge string from a density-matrix simulation, and **verify** the integration against an independent exact gate-by-gate reference.
3. **Result.** Quantify how the edge-string diagnostic scales with ansatz depth r under per-CNOT noise — the expressibility-versus-noise trade-off.

All conventions follow the existing repository: the little-endian Jordan–Wigner qubit Hamiltonian and the edge string $O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$ from `block3_core.py`.

2 Gate Errors as Quantum Channels (Theory)

2.1 The gate-then-error model

An ideal gate acts by unitary conjugation, $\mathcal{U}_g(\rho) = U_g\,\rho\,U_g^\dagger$. A noisy gate is the ideal unitary followed by an error channel $\mathcal{E}_g$,

$$\tilde{\mathcal{U}}_g = \mathcal{E}_g \circ \mathcal{U}_g, \qquad \mathcal{E}_g(\rho) = \sum_k K_k\,\rho\,K_k^\dagger, \qquad \sum_k K_k^\dagger K_k = I,$$

where the completeness relation guarantees the map is CPTP. This is exactly the model Qiskit Aer implements: for every instruction in the *transpiled* circuit, the associated error channel is applied immediately after that instruction. For a circuit $U = U_{g_M} \cdots U_{g_1}$,

$$\rho_{\text{out}} = \big(\mathcal{E}_{g_M} \circ \mathcal{U}_{g_M}\big) \circ \cdots \circ \big(\mathcal{E}_{g_1} \circ \mathcal{U}_{g_1}\big)(\rho_{\text{in}}).$$

The central consequence is that **noise scales with gate count and depth**: a deeper circuit composes more error channels.

2.2 Physical origin: from Lindblad to Pauli channels

These channels are not arbitrary. A Markovian open system obeys the Lindblad equation

$$\dot{\rho} = -i[H, \rho] + \sum_k \Big(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\}\Big),$$

and integrating over a finite gate duration τ produces a CPTP (Kraus) map. Isotropic, fully symmetric noise integrates to the *depolarizing* channel; energy relaxation $|1\rangle \to |0\rangle$ to *amplitude damping* (T_1); pure loss of phase coherence to *phase damping* (T_2). The practical Qiskit channels below are the gate-duration-integrated shadows of physical decoherence.

3 The Depolarizing Channel

The n-qubit depolarizing channel mixes the state toward the maximally mixed state. Qiskit's `depolarizing_error(p, n)` uses the "with probability p, replace by the maximally mixed state" convention,

$$\mathcal{E}_p^{\text{dep}}(\rho) = (1-p)\,\rho + p\,\frac{\operatorname{Tr}\rho}{2^n}\,I.$$

Using the Pauli twirl identity $\frac{1}{4^n}\sum_P P\rho P^\dagger = \frac{\operatorname{Tr}\rho}{2^n} I$ (sum over all 4^n Pauli strings), this is equivalently a random-Pauli channel,

$$\mathcal{E}_p^{\text{dep}}(\rho) = \Big(1 - p\,\tfrac{4^n-1}{4^n}\Big)\rho + \frac{p}{4^n}\sum_{P \neq I} P\rho P^\dagger.$$

We attach a two-qubit depolarizing error of strength p_{cx} after every `cx`, the dominant error on superconducting hardware.

4 Thermal Relaxation as a Gate Error (T_1/T_2)

Depolarizing noise is symmetric and structureless — the right first check, but not the real physics. The realistic upgrade attaches a *thermal relaxation* channel that depends on the gate duration τ and the device coherence times. Qiskit provides `thermal_relaxation_error(t1, t2, time)`, combining:

- **Amplitude damping** (T_1): excitation-loss probability $p_{T_1} = 1 - e^{-\tau/T_1}$, mapping $|1\rangle \to |0\rangle$;
- **Phase damping** (T_2): loss of off-diagonal coherence at rate set by T_2, with physical consistency $T_2 \le 2T_1$ (Qiskit enforces this).

Why this matters for the Majorana diagnostic. Two effects are qualitatively distinct from symmetric depolarizing contrast loss:

- T_1 **is parity poisoning.** In the Jordan–Wigner encoding qubit occupation tracks fermion occupation, so $|1\rangle \to |0\rangle$ flips the fermion parity and pushes the state out of the even sector the diagnostic lives in.
- T_2 **kills non-local coherence.** The edge string is an endpoint-to-endpoint correlator; pure dephasing suppresses exactly the off-diagonal coherence that carries it.

Because relaxation is gate-duration-dependent and two-qubit gates are the longest, putting the dominant noise on the `cx` operations is justified from first principles.

5 Building and Verifying a NoiseModel (Practice)

5.1 Construct, attach, read out

```
from qiskit import transpile
from qiskit.quantum_info import SparsePauliOp
from qiskit_aer import AerSimulator
from qiskit_aer.noise import (NoiseModel, depolarizing_error,
                              thermal_relaxation_error)

nm = NoiseModel()
nm.add_all_qubit_quantum_error(depolarizing_error(p_cx, 2), ['cx'])

sim = AerSimulator(method='density_matrix', noise_model=nm)
qc = ansatz.assign_parameters(theta).copy()     # the Week 7 VQE state prep
qc.save_density_matrix()
rho = sim.run(transpile(qc, sim)).result().data()['density_matrix']

O_edge = SparsePauliOp(['XZZX'], [1.0])          # L = 4 edge string
s = float(rho.expectation_value(O_edge).real)
```

`add_all_qubit_quantum_error` applies the channel after *every* occurrence of the named gate.

The transpilation gotcha. Errors attach to gate *names*, and a noise model only fires on names that survive transpilation into the simulator basis. We transpile everything to a fixed basis `['rz','sx','cx']` (`block4.py`, `BASIS_GATES`) so the `cx` the error is attached to is exactly the `cx` the circuit runs. If the target basis used `ecr` or `cz`, an error attached to `cx` would never trigger.

5.2 Precise verification: an independent exact reference

To prove the gate noise is integrated correctly — not just plausibly — we reproduce the noisy circuit a second, independent way and compare. The function `exact_reference_edge` in `block4.py` transpiles the same circuit, then evolves the density matrix *by hand*, gate by gate: it applies each gate's unitary as a `quantum_info.Operator` and, after every `cx`, applies the two-qubit depolarizing channel as a `SuperOp`. This uses none of the Aer simulator's noise machinery, so agreement is a genuine cross-check.

```
from qiskit.quantum_info import DensityMatrix, Operator, SuperOp

dep = SuperOp(depolarizing_error(p_cx, 2))
dm = DensityMatrix.from_label('0' * L)
for inst in transpile(qc, basis_gates=['rz','sx','cx'], optimization_level=0).data:
    qargs = [qc.find_bit(q).index for q in inst.qubits]
    dm = dm.evolve(Operator(inst.operation), qargs=qargs)
    if inst.operation.name == 'cx':
        dm = dm.evolve(dep, qargs=qargs)
```

Run across the whole μ sweep, the Aer `NoiseModel` edge string and this exact reference agree to $\max|\cdot| \sim 10^{-15}$ at every μ (Fig. 2, left). The gate error is therefore correctly integrated over the entire diagnostic, not merely at one density matrix.

6 Noise Integrated Through the Gates, by reps

This is the central result. The linear-entanglement `EfficientSU2` ansatz at depth r contains

$$\#\,\text{CNOTs} = r\,(L-1),$$

and each carries a depolarizing channel. The measured edge string is therefore a product of two opposing factors,

$$|\langle O_{\text{edge}}\rangle|_{\text{meas}}(r) \;\approx\; \underbrace{|\langle O_{\text{edge}}\rangle|_{\text{ideal}}(r)}_{\text{expressibility, rises with } r} \;\times\; \underbrace{(1-p_{\text{cx}})^{r(L-1)}}_{\text{gate noise, falls with } r}\;. \tag{1}$$

Week 7 established that the first factor rises toward 1 with depth (lower infidelity); Eq. (1) shows the second factor falls exponentially in depth. Their product is non-monotonic, so there is an **optimal ansatz depth** on a noisy device — the precise statement of the expressibility-versus-noise trade-off.

Plot 2 of `block4.py` makes this concrete (Fig. 1). For each depth r it prepares the VQE state across μ at fixed per-`cx` strength p_{cx}, so the only thing changing between curves is the CNOT count $r(L-1)$. As r grows, the measured edge-string sweep is pushed down and the topological/trivial contrast flattens: the same circuit depth that buys a better noiseless state costs measurable signal once the gates are noisy. An independent pure-numpy gate-by-gate evolution reproduces the same monotonic decay of purity and edge-string magnitude with CNOT count, confirming the mechanism is the accumulated channel and not a simulator artifact.

7 Verification and the Thermal Model

Plot 3 of `block4.py` combines the precise verification with the realistic noise upgrade (Fig. 2). The left panel is the exact-versus-Aer cross-check across μ described above, agreeing to $\sim 10^{-15}$. The right panel keeps the same VQE states but swaps the depolarizing channel for `thermal_relaxation_error`(T_1, T_2, τ) on each `cx`, with device-realistic coherence times $T_1 = 80\,\mu$s, $T_2 = 60\,\mu$s and a $\tau = 300$ ns gate. At these parameters the per-gate relaxation error is only $1 - e^{-\tau/T_1} \approx 0.4\%$, far below the deliberately large $p_{\text{cx}} = 0.10$ depolarizing strength, so the thermal curve sits much closer to the ideal: realistic relaxation is the *subdominant* channel on a shallow circuit, consistent with depolarizing/readout being the first bottlenecks. The thermal channel nonetheless differs in *kind* rather than only magnitude — T_1 leaks the state out of the

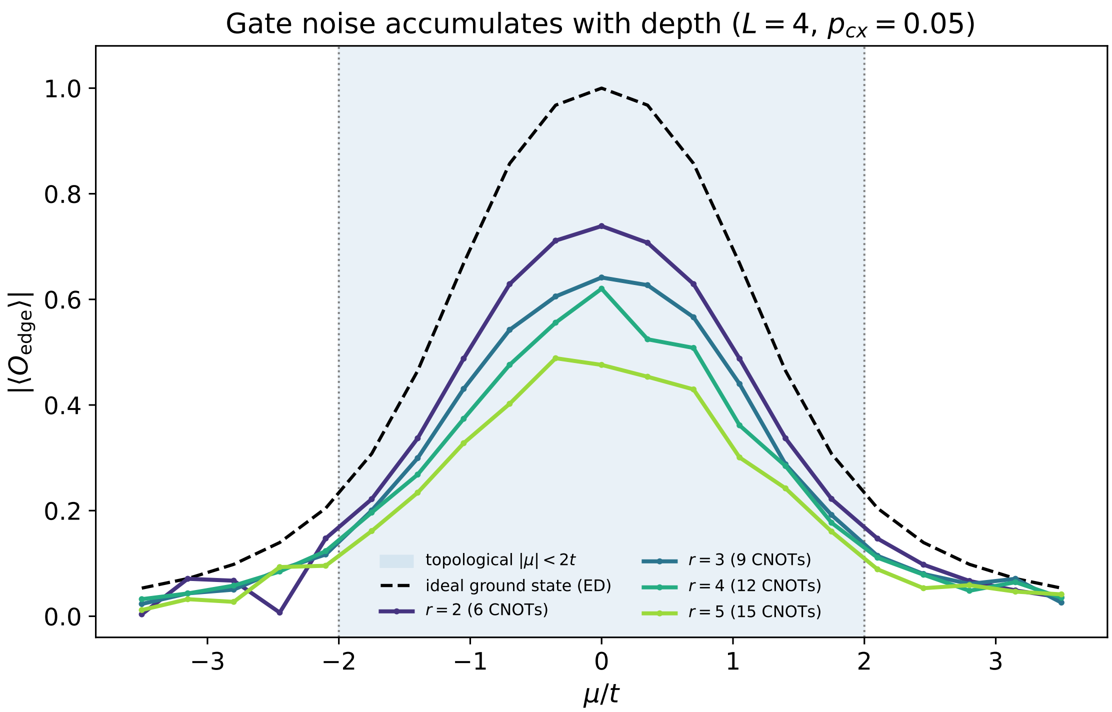

Figure 1: Edge-string sweep at increasing ansatz depth r under a fixed per-`cx` depolarizing strength. The noiseless curve (dashed) marks the topological plateau $|\mu| < 2t$; as the circuit deepens, $r(L-1)$ noisy CNOTs push the measured diagnostic down and flatten the phase contrast.

even-parity sector (parity poisoning) and T_2 attacks exactly the endpoint coherence the non-local string depends on — which is why it is the physically correct model to carry forward even when its present-parameter effect is mild.

8 Connection to the Majorana Edge-String Diagnostic

The edge string is the most exposed observable in the whole program because it is non-local: its value is a coherent correlation built across the chain by a ladder of two-qubit gates. In the circuit-level model every noisy `cx` injects depolarization that pulls the endpoint-to-endpoint coherence toward zero, suppressing $|\langle O_{\mathrm{edge}} \rangle|$ even when local observables still look reasonable; T_1 relaxation additionally poisons the parity sector, which is qualitatively worse than contrast loss; and the transition regions near $|\mu| \approx 2t$ are the most fragile, since the ideal signal is already changing rapidly there and little noise blurs the apparent phase boundary.

9 What This Enables Next

With gate errors integrated and verified at the circuit level, the next steps are concrete:

- **Noisy VQE optimization.** Optimize with a `NoiseModel`-backed Aer `Estimator`, so every cost evaluation is corrupted and the optimizer must search through noise rather than receiving a frozen validated state.

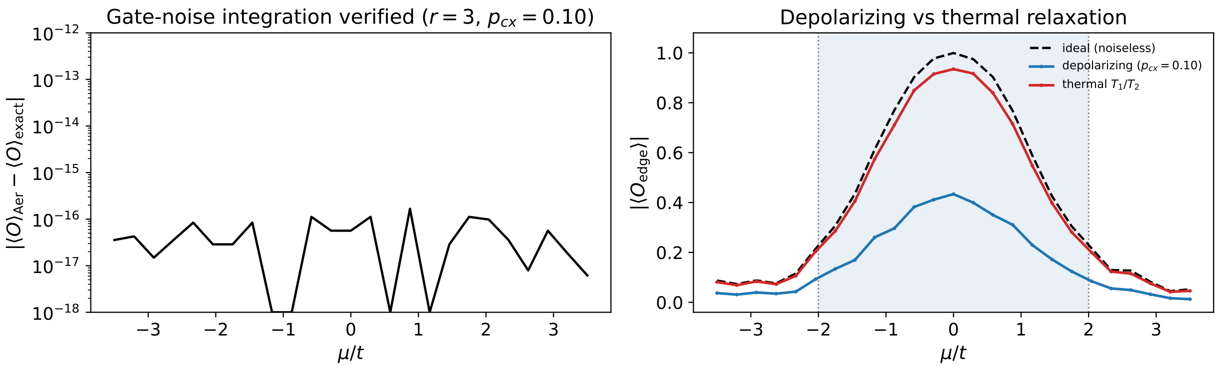

Figure 2: Left: the Aer `NoiseModel` edge string and the independent exact gate-by-gate reference agree to $\sim 10^{-15}$ across the whole μ sweep, verifying the gate-error integration. Right: depolarizing versus a physical T_1/T_2 thermal-relaxation channel at matched gate placement; the thermal model is the realistic upgrade.

- **Backend-calibrated noise.** Use `NoiseModel.from_backend(...)` for real device T_1/T_2 and gate-error data.
- **The professor's questions.** With a faithful noisy-VQE pipeline in place, return to whether the parity is protected by topology and how the chain length interacts with the depth trade-off found here.